On Erdős problem # 684

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1 Problem statement

For $0 \leq k \leq n$ write

$$\binom{n}{k} = uv$$

where the only primes dividing u are in $[2, k]$ and the only primes dividing v are in $(k, n]$. Let $f(n)$ be the smallest k such that $u > n^2$. Give bounds for $f(n)$.

2 Wandering around

Let $\binom{n}{k} = u_{n,k}v_{n,k}$, where $u_{n,k}$ is the k-smooth part of $\binom{n}{k}$ and $\mathcal{P}$ denotes the set of primes. We have:

$$u_{n,k} = \prod_{\substack{p \leq k \\ p \in \mathcal{P}}} p^{\nu_{\mathrm{p}}(\binom{n}{k})}$$

$$u_{n,k} \geq 2^{\sum_{p \leq k, p \in \mathcal{P}} \nu_{\mathrm{p}}(\binom{n}{k})}$$

Since $\nu_{\mathrm{p}}\left(\binom{n}{k}\right) \geq 1$ whenever $p \mid \binom{n}{k}$, we have:

$$\begin{aligned} u_{n,k} &\geq 2^{\sum_{p \leq k, p \in \mathcal{P}, p \mid \binom{n}{k}} 1} \\ &= 2^{\,\omega_k(\binom{n}{k})} \end{aligned}$$

where $\omega_k(\binom{n}{k})$ denotes the count of distinct prime divisors of $\binom{n}{k}$ that are less than or equal to k.

Suppose that there exists $c > 0$ such that the following hypothesis holds:

$$(H) : \omega_k(\binom{n}{k}) \geq c\,\pi(k) \quad \text{for} \quad k \sim C \ln n \ln(\ln n)$$

To ensure $u_{n,k} > n^2$, it would be sufficient to have $2^{\omega_{\leq k}(\binom{n}{k})} > n^2$. Taking the logarithm:

$$\omega_k(\binom{n}{k}) > \tfrac{2}{\ln 2} \ln n$$

Applying (H) and the Prime Number Theorem ($\pi(k) \sim \frac{k}{\ln k}$), we have this asymptotic behavior:

$$\begin{aligned} \omega_k(\binom{n}{k}) \geq c\,\pi(k) &\sim c\,\frac{k}{\ln k} \\ &\sim c\,\frac{C \ln n \ln(\ln n)}{\ln(C \ln n \ln(\ln n))} \\ &= c\,C \ln n \left(\frac{\ln(\ln n)}{\ln C + \ln(\ln n) + \ln(\ln(\ln n))}\right) \end{aligned}$$

As $n \to \infty$, the fraction in the parentheses approaches 1: $\lim_{n \to \infty} \frac{\ln(\ln n)}{\ln C + \ln(\ln n) + \ln(\ln(\ln n))} = 1$

Thus, for sufficiently large n, the condition $u_{n,k} > n^2$ is satisfied if:

$$c\,C > \frac{2}{\ln 2}$$

Therefore, choosing a proper C, if (H) holds, we obtain the improved upper bound:

$$f(n) \leq C \ln n \ln(\ln n)$$

if (H) happened to hold only for $k \sim g(n) \gg C \ln n \ln(\ln n)$, the previous approach would be fine, but would lead to poorer upper bound $f(n) \leq g(n)$
$k \sim C \ln n \ln(\ln n)$ seems to be the best growth we can hope for when using the above method, to get even better bounds, we would have to use new ways that don't make the simplification $p \to 2$ and $\nu_p(\binom{n}{k}) \to 1$ when $p \mid \binom{n}{k}$.

Conjecture 2.1 *There exists $c > 0$ such that,*

$$\omega_k(\tbinom{n}{k}) \geq c\,\pi(k) \quad \textit{for} \quad k \sim C \ln n \ln(\ln n)$$

for $\omega_k(\binom{n}{k})$ the count of distinct prime divisors of $\binom{n}{k}$ under k

Proof.
We first wanted to see, how $\omega_k(\binom{n}{k})$ would behave in practice. Mainly to reassure that the lemma could hold an thus be proven, but also to try to get insight that could help us derive the proof. For example if the ratio converged to 0, this would have been quite devastating for the lemma, if it converged to 1, it would imply a different behavior that a convergence to 1/2.

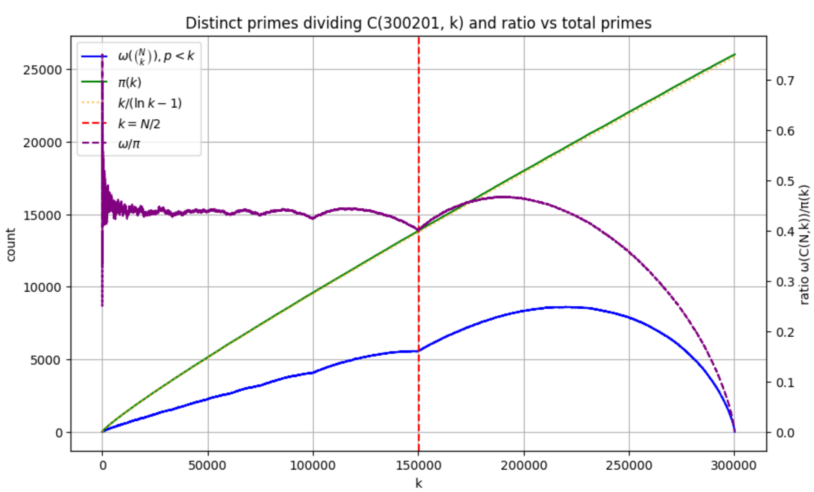

Figure 1: for a randomly chosen "large" n, here $n = 300201$, we plot the ratio between $\pi(k)$ and the amount of distinct prime divisors under k : ω_k of $\binom{n}{k}$, it seems to plateau a bit above 0.4 (we only care about the behavior before $n/2$), this behavior appears quite similar for all the different n we tried.

This is quite intriguing, as the ratio seems to converge to a value that is a bit above 0.4 and seems a bit arbitrary. But also quite reassuring in that the lemma could be proven to hold for large n.

Let's try to prove the lemma now

$$\begin{aligned}
\omega_k(\tbinom{n}{k}) &= \sum_{p \le k,\, p \in \mathcal{P}} \mathbb{1}_{p | \binom{n}{k}} \\
&\underset{??}{=} \sum_{p \le k,\, p \in \mathcal{P}} \left\{ \frac{k}{p} \right\} \;+\; o(\pi(k)) \\
&= (1-\gamma)\,\pi(k) \;+\; o(\pi(k))
\end{aligned}$$

Indeed from F.Pillichshammer [Pil10] and the corresponding stack post, we have, as $x \to \infty$:

$$\sum_{p \le x,\, p \in \mathcal{P}} \left\{ \frac{x}{p} \right\} = \frac{x}{\ln x}(1-\gamma) + o(\frac{x}{\ln x})$$

Thus, $\sum_{p \le k,\, p \in \mathcal{P}} \left\{ \frac{k}{p} \right\} = (1-\gamma)\,\pi(k) + o(\pi(k))$

According to [Pil10], it also remains true if we sum over all divisors that are prime powers instead of primes. More precisely, we have : $\sum_{p^v \le k,\, p \in \mathcal{P}} \left\{ \frac{k}{p^v} \right\} = (1-\gamma)\,\pi(k) + o(\pi(k))$

The only part we are missing now is to prove the following (or something close):

$$\omega_k(\tbinom{n}{k}) = \sum_{p \le k,\, p \in \mathcal{P}} \left\{ \frac{k}{p} \right\} \;+\; o(\pi(k))$$

for $\omega_k(\binom{n}{k})$ the count of distinct prime divisors of $\binom{n}{k}$ under k, and $k \sim \ln(n)\ln(\ln(n))$

If we have this, then we would have the following upper-bound: $f(n) \le C\, \ln(n)\, \ln(\ln(n))$

The $f(n) \le C\, \ln(n)\, \ln(\ln(n))$ seems to reflect the growth obtain empirically quite well:

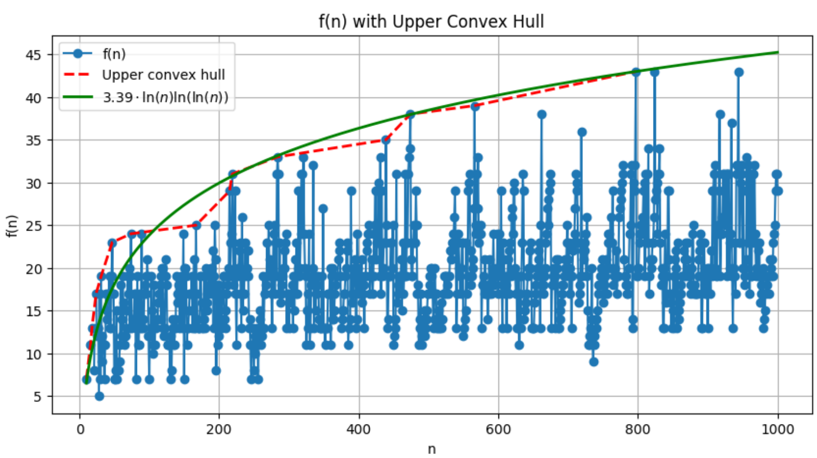

Figure 2: We plot $f(n)$, its upper hull and a $C\ln(n)\ln(\ln(n))$ growth.

Let's prove first that

$$\boxed{\omega_k(\binom{n}{k}) := \sum_{p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \underset{??}{=} \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{(\{\frac{n}{p}\}>\{\frac{k}{p}\})} \;+\; o(\pi(k))}$$

proof. First, we can restrain to $\sqrt{k}\leq p\leq k$, indeed, we have:

$$\begin{aligned}
\omega_k(\binom{n}{k}) &:= \sum_{p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \\
&= \sum_{p\leq \sqrt{k},\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \;+\; \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \\
&= O(\pi(\sqrt{k})) + \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \\
&= \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \;+\; o(\pi(k))
\end{aligned}$$

Now let's prove that for $\sqrt{k}\leq p\leq k$, we have $p \mid \binom{n}{k} \Leftrightarrow \{\frac{n}{p}\} < \{\frac{k}{p}\}$.

For $p>\sqrt{k}$, $p^2>k$, we thus have

$$\nu_p(\binom{n}{k}) = \lfloor\frac{n}{p}\rfloor - \lfloor\frac{n-k}{p}\rfloor - \lfloor\frac{k}{p}\rfloor$$

and $\lfloor x\rfloor + \lfloor y\rfloor = \lfloor x+y\rfloor$ iff $\{x\}+\{y\}<1$. So, in the case $\{\frac{n}{p}\} < \{\frac{k}{p}\}$:
$\{\frac{n-k}{p}\} = \{\frac{n}{p}\} - \{\frac{k}{p}\}$, so $\{\frac{k}{p}\} + \{\frac{n-k}{p}\} = \{\frac{n}{p}\} < 1$ Therefore, $\nu_p(\binom{n}{k}) = 0$

and equivalently, $\lfloor x+y\rfloor = \lfloor x\rfloor + \lfloor y\rfloor + 1$ iff $\{x\}+\{y\}\geq 1$. So, in the case $\{\frac{n}{p}\} \geq \{\frac{k}{p}\}$:
$\{\frac{n-k}{p}\} = \{\frac{n}{p}\} - \{\frac{k}{p}\} + 1$, so $\{\frac{k}{p}\} + \{\frac{n-k}{p}\} = 1 + \{\frac{n}{p}\} \geq 1$ Therefore, $\nu_p(\binom{n}{k}) \geq 1$
We have $p \mid \binom{n}{k} \Leftrightarrow \{\frac{n}{p}\} < \{\frac{k}{p}\}$

Thus

$$\omega_k(\binom{n}{k}) = \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \mathbb{1}_{(\{\frac{n}{p}\}>\{\frac{k}{p}\})} \;+\; o(\pi(k))$$

Now, can we restrain the other side to $\sqrt{k}\leq p\leq k$? to have:

$$\boxed{\sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} = (1-\gamma)\,\pi(k) \;+\; o(\pi(k))}$$

as $\{\frac{k}{p}\}\leq 1$, $\sum_{p\leq\sqrt{k},\, p\in\mathcal{P}} \{\frac{k}{p}\} = O(\pi(\sqrt{k}))$ Thus we have the following:

$$\begin{aligned}
\sum_{p\leq k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} &= (1-\gamma)\,\pi(k) \;+\; o(\pi(k)) \\
\sum_{p\leq\sqrt{k},\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} \;+\; \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} &= (1-\gamma)\,\pi(k) \;+\; o(\pi(k)) \\
O(\pi(\sqrt{k})) \;+\; \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} &= (1-\gamma)\,\pi(k) \;+\; o(\pi(k)) \\
o(\pi(k)) \;+\; \sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} &= (1-\gamma)\,\pi(k) \;+\; o(\pi(k)) \\
\sum_{\sqrt{k}\leq p\leq k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} &= (1-\gamma)\,\pi(k) \;+\; o(\pi(k))
\end{aligned}$$

So, to summarize:

$$\begin{aligned}
\omega_k(\binom{n}{k}) := & \sum_{p\le k,\, p\in\mathcal{P}} \mathbb{1}_{p|\binom{n}{k}} \\
= & \sum_{\sqrt{k}\le p\le k,\, p\in\mathcal{P}} \mathbb{1}_{\left(\{\frac{n}{p}\}>\{\frac{k}{p}\}\right)} \;+\; o(\pi(k)) \\
\underset{??}{=} & \sum_{\sqrt{k}\le p\le k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} \;+\; o(\pi(k)) \\
= & \quad (1-\gamma)\,\pi(k) \;+\; o(\pi(k))
\end{aligned}$$

To conclude, and have the $C\ln(n)\ln\ln(n)$ upper-bound, we are thus just missing the following:

Conjecture 2.2 *We have, for $k \sim C\ln(n)\ln(\ln(n))$:*

$$\sum_{\sqrt{k}\le p\le k,\, p\in\mathcal{P}} \mathbb{1}_{\left(\{\frac{n}{p}\}>\{\frac{k}{p}\}\right)} \quad \underset{??}{=} \sum_{\sqrt{k}\le p\le k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} \;+\; o(\pi(k))$$

This seems doable, as if $0 \le \left\{\frac{n}{p}\right\} < 1$ is uniformly distributed on $[0,1]$, $\mathbb{P}\Big(\left\{\frac{n}{p}\right\} < \left\{\frac{k}{p}\right\}\Big) = \left\{\frac{k}{p}\right\}$

we would also have the previous conjecture verified, if we directly proved it for $2 \le p \le k$:

$$\sum_{p\le k,\, p\in\mathcal{P}} \mathbb{1}_{\left(\{\frac{n}{p}\}>\{\frac{k}{p}\}\right)} \quad \underset{??}{=} \sum_{p\le k,\, p\in\mathcal{P}} \left\{\frac{k}{p}\right\} \;+\; o(\pi(k))$$

If we verify the previous conjecture, it would verify the first one, and let us conclude on the veracity of the $f(n) \le C\ln(n)\ln\ln(n)$ upper-bound.

References

[Pil10] Friedrich Pillichshammer. Euler's constant and averages of fractional parts. *The American Mathematical Monthly*, 117(1):78–83, 2010.